

SYDNEY TECHNICAL HIGH SCHOOL
(Est 1911)

MATHEMATICS EXTENSION I

YEAR 11 COMMON TEST

JULY 2002

Time allowed : 70 minutes

Instructions :

- Show all necessary working in every question.
- Start each question on a new page.
- Attempt all questions.
- All questions are not of equal value.
- Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- These questions are to be handed in with your answers.

Name : _____

Class : _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total

Question 1

Marks

a) Write the expansion of $\tan(A + B)$

1

b) Let A be the point $(6, -12)$ and let B be the point $(-4, 8)$.

2

Find the coordinates of the point P which divides the
interval AB internally in the ratio $5 : 3$.

c) Evaluate $\lim_{x \rightarrow \infty} \frac{2x^2 - x - 3}{x^2 + 4x + 8}$

1

d) Express $3 \sin \theta$ in terms of t where $t = \tan \frac{\theta}{2}$.

1

e) Express as a single trigonometric ratio

1

$$\sin \alpha \cos(\alpha - \beta) - \cos \alpha \sin(\alpha - \beta)$$

f) Find the point on the curve $y = x^2 - 8x + 4$ where its tangent

2

is parallel to the line $2x + y + 6 = 0$.

Question 2 (Start a new page)a) Evaluate $\lim_{x \rightarrow 3} \frac{x^2 + 2x - 15}{x^2 - 9}$

2

b) Show that $\frac{\sin x}{1 + \cos x} + \frac{1 + \cos x}{\sin x} = 2 \operatorname{cosec} x$

3

c) Use the formula $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

3

to find the derivative of $f(x) = 4x^2 - x$.

Question 3 (Start a new page)

- a) Find to the nearest degree the acute angle between the lines 3

$$4x - y - 3 = 0 \quad \text{and} \quad x - 3y - 6 = 0.$$

- b) i) Show that $\cos 3A = 4\cos^3 A - 3\cos A$. 3

- ii) Hence solve the equation

$$4\cos^3 A - 3\cos A = \frac{1}{2} \quad \text{for } 0^\circ \leq A \leq 180^\circ \quad 3$$

Question 4 (Start a new page)

- a) Find the equation of the normal to the curve $y = x^4 - \frac{3}{x}$ 3

at the point $(1, -2)$.

- b) Solve $2\sin^2 \theta \sec \theta - \tan \theta = 0$ for $0 \leq \theta \leq 2\pi$ (answer in radians) 3

- c) Find the value, or values of k given that the perpendicular distance 3

of the point $(3, k)$ from the line $2x - y + 4 = 0$ is equal to $2\sqrt{5}$ units.

Question 5 (Start a new page)

a) Differentiate $y = \frac{3x^2}{x-1}$ 2

b) Find the equation of the line which passes through the point 3
of intersection of the lines $x + 4y - 8 = 0$ and $5x + 6y - 6 = 0$
and the point $(4, -1)$. Give your answer in general form.

c) Solve to the nearest degree the equation 3

$$2\cos 2x - 7\cos x = 0 \text{ for } 0^\circ \leq x \leq 360^\circ.$$

Question 6 (Start a new page)

a) Find the acute angle between the line AB and the line BC given 3
given that A , B and C have the coordinates

$(-3, 4)$, $(2, 1)$ and $(2, 6)$ respectively.

b) If $y = (2x - 1)\sqrt{4x - 1}$ 3

show that $\frac{dy}{dx} = \frac{12x - 4}{\sqrt{4x - 1}}$

c) Solve the equation $3\cos\theta - \sin\theta = 2$ for $0^\circ \leq \theta \leq 360^\circ$ 3
giving your answer correct to the nearest degree.

Solutions

Question 1

$$a. \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$b. A(6, -4) \quad B(-3, 8) \quad S(3)$$

$$\left(\frac{3 \times 6 + 8 \times -4}{8}, \frac{3 \times -4 + 8 \times 8}{8} \right)$$

$$= \left(-\frac{1}{2}, \frac{5}{2} \right)$$

$$\lim_{x \rightarrow \infty} \frac{2x^2 - x - 3}{x^2 + 4x + 8} = 2$$

$$d. 3 \sin \theta = 3 \times \frac{2t}{1+t^2}$$

$$= \frac{6t}{1+t^2}$$

$$e. \sin x \cos(x-p) - \cos x \sin(x-p)$$

$$= \sin(x - (x-p))$$

$$= \sin p$$

$$f. 2x + y + 6 = 0 \quad m = -2$$

$$\frac{dy}{dx} = 2x - 8$$

$$\therefore 2x - 8 = -2$$

$$2x = 6$$

$$x = 3$$

$$\therefore y = 3^2 - 8 \times 3 + 4$$

$$= -11$$

$$\therefore \text{at point } (3, -11)$$

Question 2

$$a. \lim_{x \rightarrow 3} \frac{(x+5)(x-2)}{(x-3)(x-3)}$$

$$= \lim_{x \rightarrow 3} \frac{x-3}{x+3}$$

$$= \frac{3}{6}$$

$$= \frac{1}{2}$$

$$b. LHS = \frac{\sin x}{1 + \cos x} + \frac{1 + \cos x}{\sin x}$$

$$= \frac{\sin^2 x + (1 + \cos x)^2}{(1 + \cos x) \sin x}$$

$$= \frac{\sin^2 x + \cos^2 x + 2(\cos x + 1)}{(1 + \cos x) \sin x}$$

$$= \frac{2 + 2 \cos x}{(1 + \cos x) \sin x}$$

$$= \frac{2(1 + \cos x)}{(1 + \cos x) \sin x}$$

$$= \frac{2}{\sin x}$$

$$= 2 \operatorname{cosec} x$$

$$= RHS$$

$$c. \frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{[4(x+h)^2 - (x+h)] - [4x^2 - x]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{4x^2 + 8xh + 4h^2 - x - h - 4x^2 + x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{8xh + 4h^2 - h}{h}$$

$$= \lim_{h \rightarrow 0} 8x + 4h - 1$$

$$= 8x - 1$$

Question 3

a. $m_1 = 4$, $m_2 = 3$

$$\therefore \tan \theta = \left| \frac{\frac{1}{3} - 4}{1 + 4 \cdot \frac{1}{3}} \right|$$

$$= \frac{11}{7}$$

$$\therefore \theta = 58^\circ$$

b. i) LHS = $\cos 3A$

$$= \cos(A+2A)$$

$$= \cos A \cos 2A - \sin A \sin 2A$$

$$= \cos A (2\cos^2 A - 1) - \sin A (2\sin A \cos A)$$

$$= 2\cos^3 A - \cos A - (1 - \cos^2 A) 2\cos A$$

$$= 2\cos^3 A - \cos A - 2\cos A + 2\cos^3 A$$

$$= 4\cos^3 A - 3\cos A$$

$$= \text{RHS}$$

ii) $4\cos^3 A - 3\cos A = \frac{1}{2}$

$$\cos 3A = \frac{1}{2}$$

$$3A = 60^\circ, 300^\circ, 420^\circ$$

$$A = 20^\circ, 100^\circ, 140^\circ$$

Question 4

a. $y = x^4 - 3x^{-1}$

$$\frac{dy}{dx} = 4x^3 + 3x^{-2}$$

$$= 4x^3 + \frac{3}{x^2}$$

when $x = 1$

$$m_T = 4 + 3$$

$$= 7$$

$$\therefore m_N = -\frac{1}{7}$$

$$\therefore \text{equation } y - y_1 = m(x - x_1)$$

$$y + 2 = -\frac{1}{7}(x - 1)$$

$$7y + 14 = -x + 1$$

$$x + 7y + 13 = 0$$

b. $2\sin^2 \theta \sec \theta - \tan \theta = 0$

$$\frac{2\sin^2 \theta}{\cos \theta} - \frac{\sin \theta}{\cos \theta} = 0$$

$$2\sin^2 \theta - \sin \theta = 0$$

$$\sin \theta (2\sin \theta - 1) = 0$$

$$\sin \theta = 0 \quad \sin \theta = \frac{1}{2}$$

$$\theta = 0, \pi, 2\pi, \frac{\pi}{6}, \frac{5\pi}{6}$$

c. $\frac{|2 \times 3 + -1 \times k + 4|}{\sqrt{2^2 + 1^2}} = 2\sqrt{5}$

$$\frac{|10 - k|}{\sqrt{5}} = 2\sqrt{5}$$

$$|10 - k| = 10$$

$$10 - k = 10 \quad \text{or} \quad 10 - k = -10$$

$$k = 0 \quad \text{or} \quad k = 20$$

Question 5

a. $\frac{dy}{dx} = \frac{(x-1)(x - 3x^2(1))}{(x-1)^2}$

$$= \frac{6x^2 - 6x - 3x^2}{(x-1)^2}$$

$$= \frac{3x^2 - 6x}{(x-1)^2}$$

$$b. \quad 2x + 4y - 5 + k(5x + 3y - 6) = 0$$

$$\text{e.g. } (4, -1)$$

$$4 - 4 - 5 + k(20 - 6 - 6) = 0$$

$$-8 + 8k = 0$$

$$k = 1$$

$$6x + 10y - 14 = 0$$

$$3x + 5y - 7 = 0$$

$$c. \quad 2 \cos 2x - 7 \cos x = 0$$

$$2(2 \cos^2 x - 1) - 7 \cos x = 0$$

$$4 \cos^2 x - 7 \cos x - 2 = 0$$

$$(4 \cos x + 1)(\cos x - 2) = 0$$

$$\cos x = -\frac{1}{4} \quad \cos x = 2$$

no solution

$$x = 104^\circ, 256^\circ$$

$$b. \quad y = (2x-1)\sqrt{4x-1}$$

$$y = (2x-1)(4x-1)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = 2(4x-1)^{\frac{1}{2}} + (2x-1)^{\frac{1}{2}} \cdot (4x-1)^{-\frac{1}{2}} \cdot 4$$

$$= 2\sqrt{4x-1} + \frac{2(2x-1)}{\sqrt{4x-1}}$$

$$= \frac{2(4x-1) + 2(2x-1)}{\sqrt{4x-1}}$$

$$= \frac{12x-4}{\sqrt{4x-1}}$$

$$c. \quad 3 \cos \theta - \sin \theta = 2$$

$$\sqrt{10} \cos(\theta + \alpha) = 2$$

$$\tan \alpha = \frac{1}{3}$$

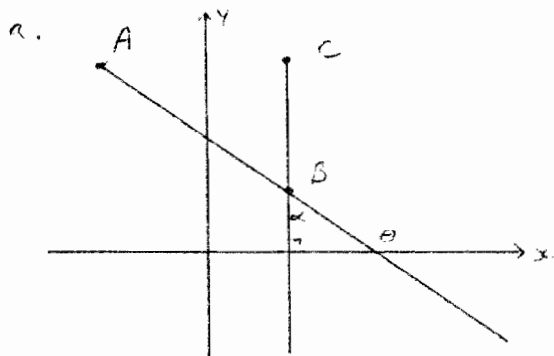
$$\alpha = 18^\circ 26'$$

$$\therefore \cos(\theta + \alpha) = \frac{2}{\sqrt{10}}$$

$$\theta + \alpha = 50^\circ 46', 309^\circ 14'$$

$$\theta = 32^\circ, 291^\circ$$

QUESTION 6



$$\tan \theta = m_{AB}$$

$$= \frac{4-1}{-3-2}$$

$$= -\frac{3}{5}$$

$$\therefore \theta = 147^\circ$$

$$\therefore \alpha = 59^\circ$$

$$\therefore \text{angle is } 59^\circ$$